CAN Debugging

While CAN is an especially reliable method of communication due to the twisted wires, there are many ways that CAN can go wrong in setup.

The following is a list of potential problems that may be encountered:

-Inconsistent or incorrect signals

-No signals visible on receiving device

Inconsistent or Incorrect Signals

Inconsistent or incorrect signals often come from noise on the CAN bus. An easy way to check for noise on the CAN bus is to connect an oscilloscope as a CAN device, connecting one probe to the CANH line, and one to the CANL, (see ***Figure 1***). If the oscilloscope shows a clear square wave,(***Figure 2***), then the issue lies with the receiving device. The following are possible causes:

-Poor wiring connection between transmitter and transceiver, (a poor connection between transceiver and Teensy can result in inconsistent signals)

-CAN lines saturated with messages, (if a device on the CAN line is sending too many messages, other messages may not be able to be sent. This can be verified via CAN sniffer, (**Figure 1**), or by echoing messages received between two microcontrollers)

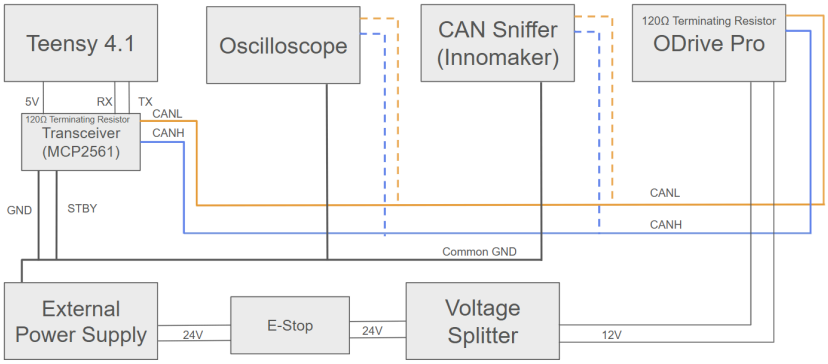


Figure 1. Oscilloscope and CAN Sniffer Connected to CANH and CANL

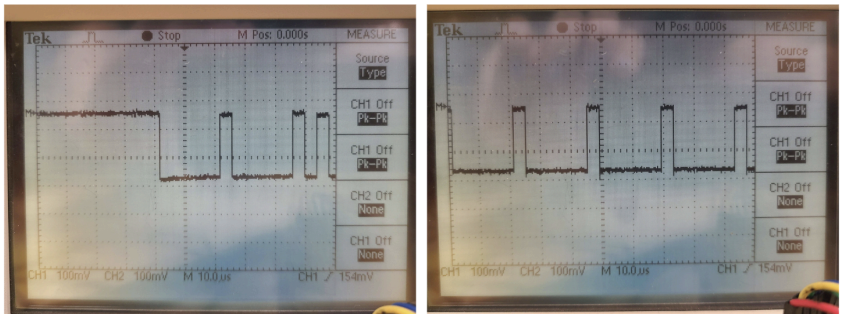


Figure 2. Healthy Heartbeat Signals on an Oscilloscope

If the oscilloscope shows a large amount of noise that exceeds the magnitude of the square wave, the following are possible causes:

-High electromagnetic interference, (try isolating the wires, or twisting them tighter to see if that reduces noise in the signal)

-Incorrect terminating resistors, (if the resistance measured from any CANH to CANL does not show 60Ohms, the two 120Ohm resistors are incorrect)

-Poor ground connections, (verify that the grounds on each CAN bus have no voltage difference between them)

-Incorrect voltage to CAN bus, (verify that each CAN bus is receiving the specified amount of voltage. See *Wiring and Power* for details on power)

No Signals Visible on the Receiving Device

When no signals are present on the receiving device, it can be due to either a software or hardware issue. This can usually be determined by using an oscilloscope as shown in **Figure 1** to determine if a signal is present on the lines at all. If there is a clear square wave, the issue lies with the software or configurations. If there is no square wave, or a high noise wave, the issue lies with hardware.

Clear Square Wave:

-Node IDs incorrect, (if a node ID is unconfigured, that device will not send or receive any messages. In addition, if a receiving device is looking for an incorrect node ID, it will not recognize the signal on the CAN lines)

-Mismatched baudrates, (if the baudrates of the sending and receiving devices are not matching, no messages will be picked up the receiving device)

-Extended message IDs, (An unlikely, but possible, error is having a device configured to use different message formats than the other devices. The ODrive had a setting for extended message IDs, which changes the way the node IDs are labelled. This is disabled by default, but if messages are present on a CAN sniffer with unusual node IDs, this could be a cause)

-Complete saturation of CAN messages, (an issue encountered in this project was the Teensy fully saturating the CAN lines with error messages. Something, (likely an incorrect voltage on the transmitting lines), caused the Teensy to collect error messages until it hit the error limit, (usually 127 error messages). The Teensy then saturated the CAN busses with “STATE_ERROR_PASSIVE”, preventing any other message from being transmitted. Clearing the errors solved this issue, and the errors did not regenerate)

-Faulty receiving device, (another issue encountered in this project was the Teensy transceiver having a poor physical connection with the Teensy. This was verified by measuring the resistance between the Teensy RX pin and the transceiver RX pin. Ideal resistance should be very low, < 0.5Ohms. Signal quality can also be verified by using a single oscilloscope probe on the RX line of the Teensy. It should show a clear square wave if the Teensy is physically receiving the messages from CANH and CANL)

Unclear or Absent Square Wave:

For high noise square waves, if none of the solutions for inconsistent signals work, the issue may lie with the hardware itself. Checking parts one at a time, and swapping faulty parts may

narrow down the issue. If there is a complete absence of a wave, no device is sending signals. This is either due to a hardware issue with the CAN bus on a device, or the device is not configured to send any signals, heartbeat or otherwise.